

SEEDPRINTS: FINGERPRINTS CAN EVEN TELL WHICH SEED YOUR LARGE LANGUAGE MODEL WAS TRAINED FROM

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ABSTRACT

Fingerprinting Large Language Models (LLMs) is increasingly indispensable for provenance verification, ownership auditing, and model attribution. Existing fingerprinting techniques, however, are predominantly assessed only after fine-tuning, at which point models have already accumulated comparatively stable signatures from training data, optimization trajectories, and hyperparameter choices. Because the overwhelming majority of a model’s capacity and knowledge is acquired during pretraining rather than during downstream adaptation, large-scale pretraining constitutes the more fundamental regime in which lineage verification must operate. We demonstrate that existing fingerprinting methods become *unreliable* in this regime, since they depend on post-hoc signatures that materialize only after substantial training; this limitation is at odds with the classical Galtonian conception of a fingerprint as an intrinsic and persistent identity marker. Against this backdrop, we formulate a stronger and more intrinsic notion of LLM fingerprinting: **SeedPrints**, a method that exploits random-initialization biases as persistent, seed-dependent identifiers that already exist before training begins. We show that untrained models display reproducible prediction biases induced by their initialization seed, and that these initially weak signals remain statistically detectable throughout training, thereby enabling high-confidence lineage verification. Whereas prior techniques fail during early pretraining or deteriorate under distributional shifts, **SeedPrints** remains effective across the entire training process, from initialization through large-scale pretraining and downstream adaptation. Experiments on LLaMA-style and Qwen-style models establish seed-level distinguishability and support birth-to-lifecycle identity verification. Evaluations on large-scale pretraining trajectories and real-world fingerprinting benchmarks further substantiate robustness under prolonged optimization, domain shift, and parameter-altering transformations. Collectively, our findings indicate that initialization itself imprints a unique and durable identity on LLMs, thereby constituting a genuinely “Galtonian” fingerprint. Code will be released upon acceptance.

1 INTRODUCTION

LLM fingerprints have recently emerged as behavioral instruments for identifying, attributing, and tracing large language models through their observable responses (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024). By establishing a verifiable association between a suspicious model and its purported source, these methods are intended to help model owners detect theft, unauthorized reuse, and other forms of illicit model appropriation (Yoon et al., 2025; Zhang, 2025).

Much of this literature explicitly derives its terminology from Francis Galton’s biological fingerprint metaphor in *Finger Prints* (1892) (Galton, 1892):

“A fingerprint is the pattern formed by friction-ridge skin on the fingertips; this ridge configuration is individually unique and essentially permanent across an individual’s lifetime.”

SeedPrints vs Baselines: OLMo-2-7B Pre-training Checkpoints

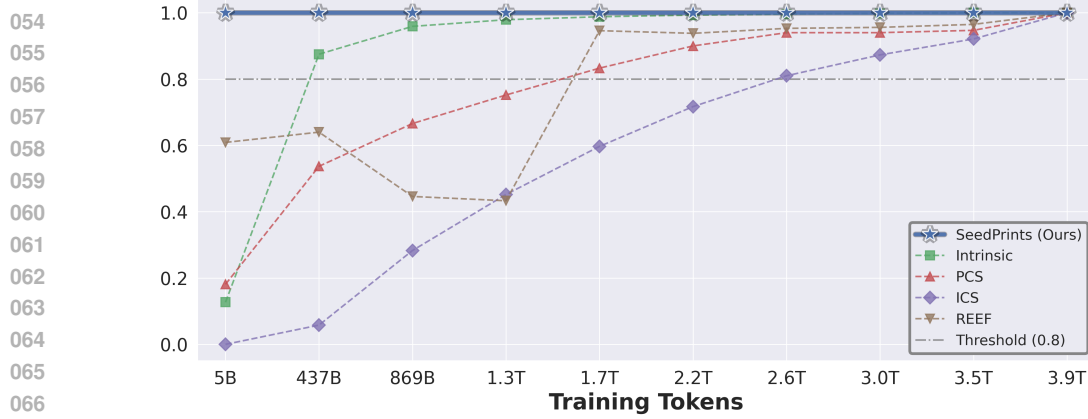


Figure 1: **Existing fingerprinting methods fail to detect model lineage during early pre-training.** We evaluate five methods on OLMo-2-7B checkpoints spanning 5B to 3.9T training tokens, with every checkpoint compared against the final checkpoint. The y-axis reports the similarity score, where larger values indicate stronger lineage evidence, and the dashed horizontal line denotes the 0.8 detection threshold. Although all baselines deteriorate and fall below the threshold for early checkpoints, **SeedPrints** attains perfect detection ($p \ll 0.001$, visualized as $1 - p$) from the first checkpoint onward.

The analogy imposes a stringent desideratum: an effective fingerprint should be simultaneously unique and persistent, and should be present from the moment of a model’s “birth” at initialization. Most existing methods described as fingerprinting, however, do not satisfy this criterion (Pasquini et al., 2024; Xu et al., 2024; Yoon et al., 2025; Zhang et al., 2024; Zeng et al., 2024; Zhang, 2025; Luan et al., 2025; Tsai et al., 2025; Alhazbi et al., 2025). As illustrated in Figure 1, they cannot reliably verify lineage during early pretraining, largely because their signatures are defined only after models have effectively converged, for example by extracting patterns from parameters or generated text. Consequently, the separability they obtain reflects not an intrinsic birthmark of the model, but rather the downstream imprint of training data, optimization dynamics, and hyperparameter choices. In Section 5.1, for instance, existing baselines frequently fail to recover lineage when the data distribution changes substantially. Such methods therefore operate more as post hoc identifiers than as **Galtonian fingerprints**: innate and immutable marks that accompany a model from its inception.

In this work, we advance a stricter notion of LLM fingerprinting, namely an intrinsic property that is already present at *initialization* and remains detectable at *any* subsequent point in training. Our principal contributions are as follows:

- **Identifying a fundamental limitation of existing methods.** We show that prior fingerprinting approaches fail to verify lineage reliably during early pretraining and can be readily confounded by substantial shifts in the training-data distribution. (Section 5.1)
- **Discovering and leveraging initialization-driven fingerprints.** We reveal that untrained models exhibit seed-dependent biases in their internal representations, which constitute weak but persistent identity signals already present at initialization. (Section 3) We then introduce **SeedPrints**, a method that isolates these initialization-born fingerprints while remaining substantially more robust to confounding factors such as data distribution, training duration, and optimization dynamics. (Section 4)
- **Extensive empirical validation.** We demonstrate that our method distinguishes models that differ only in initialization seed, even under identical training pipelines and data orders, and that the resulting signal persists under continual training across diverse datasets. (Section 5.1) We further show that **SeedPrints** remains effective across all training stages, including large-scale pretraining trajectories and realistic downstream deployment scenarios on the *LeaFBench* benchmark (Shao et al., 2025), involving instruction tuning, fine-tuning, PEFT, quantization, merging, and distillation. (Sections 5.2 and 5.3)

2 RELATED WORK

LLM protection methods can be organized into two broad families: (i) *watermarking / active fingerprinting* approaches, which *insert* an identifiable signature into a model or into its outputs (Xu et al., 2024; Nasery et al.; Tsai et al., 2025; Sander et al., 2024), and (ii) *passive fingerprinting* approaches, which *extract* a signature from a model’s pre-existing behavior without modifying the model itself (Yoon et al., 2025; Zhang et al., 2024; Alhazbi et al., 2025; Pasquini et al., 2024; Suzuki et al., 2025; Zhang, 2025).

Watermarking and Active Fingerprinting. Active approaches deliberately implant verifiable identifiers that can later support ownership checks. Classical instantiations include backdoor-based mechanisms (Adi et al., 2018; Li et al., 2019; Zhang et al., 2018), digital signatures, and hash-function-based constructions (Guo & Potkonjak, 2018; Zhu et al., 2020). For language models, two common variants are *text watermarks*, which bias generation or introduce predefined patterns to encode hidden information (Kirchenbauer et al., 2023; Xu et al., 2024; Nasery et al.), and *model-weight watermarks*, which embed identifiers into parameters or associate them with secret triggers through fine-tuning (Luan et al., 2025; Li et al., 2023). Although backdoor-style fingerprints are operationally simple and can survive moderate fine-tuning (e.g., Dasgupta et al. 2024), their invasive character presupposes control over the training process, making them unsuitable for retroactively marking third-party models.

Passive LLM Fingerprinting. Passive fingerprinting, by contrast, identifies models by analyzing intrinsic properties without imposing any modification. These techniques differ according to the assumed access level. Under **white-box access**, signatures are extracted from model weights, exploiting properties such as attention-matrix distributional profiles (Yoon et al., 2025), centered-kernel-alignment similarity among internal representations (Zhang et al., 2024), or stable directions in parameter vectors. Under **black-box access**, fingerprinting relies on input-output behavior, using crafted queries (Pasquini et al., 2024), distinctive prompt-response pairs (Tsai et al., 2025), stylometric regularities (Alhazbi et al., 2025), or iterative prompting–response games (Iourovitski et al., 2024), although such methods may be less robust to fine-tuning. Nevertheless, these methods define their signatures *post hoc*: they identify properties that emerge from a completed training process rather than the innate, “Galtonian” fingerprints present at random initialization that our work seeks to characterize. Consequently, *they are not designed for lineage verification during large-scale pretraining*, precisely the regime in which post-training signatures have not yet formed.

3 BIASES ORIGINATING FROM INITIALIZATION PERSIST AFTER TRAINING

In this section, we develop the central empirical observation underlying our method: language models exhibit pronounced prediction biases that originate in the random initialization seed, and these biases, although attenuated by optimization, remain detectable after training.

Initialization induces seed-specific bias profiles. We evaluate a LLaMA-2-style model initialized with seed 123 on 10,000 random input sequences of length 1,024, with tokens sampled uniformly from the vocabulary. In Figure 2 (Left), we report the frequency with which each output token (upper panel, final logits) and each hidden dimension (lower panel, last-layer representations) attains the minimum value across the 10,000 random trials. The strongly non-uniform morphology of both distributions indicates the presence of extreme output-preference biases. When the experiment is repeated across initialization seeds, the global magnitude of the bias remains broadly stable, yet the specific argmin dimensions vary with the seed (Figure 2, Lower Right). These observations accord with recent evidence that biases in randomly initialized models arise from inter-sequence representation contraction, a phenomenon driven by asymmetric nonlinear activations in MLP blocks and further amplified by self-attention (Li et al., 2026). From this perspective, distinct initialization seeds induce distinct contraction directions, which in turn produce seed-specific argmin patterns across output dimensions.

Training preserves initialization-born bias. Although training substantially alters output magnitudes, it does not completely erase the relative preference over output dimensions established at initialization. We train the previously initialized model (seed 123) for one epoch on OpenWebText (Gokaslan et al., 2019) and evaluate intermediate checkpoints. Focusing on the $m = 50$ output dimensions most frequently assigned minimum values by the initialized model, we compute, for each checkpoint, the correlation between checkpoint responses and initialized-model responses across random inputs restricted to these dimensions; operationally, this statistic measures how similarly the

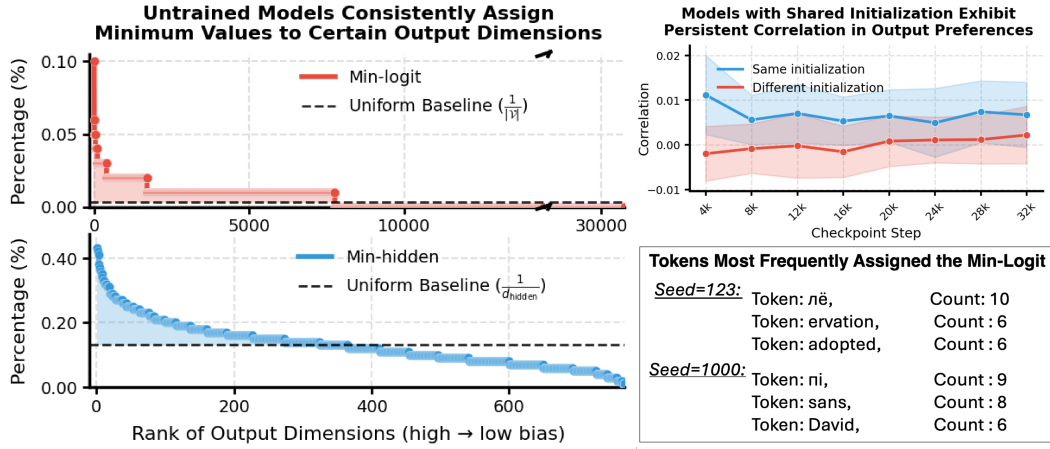


Figure 2: **Initialization-born output bias persists through training.** *Left:* Under completely random inputs, the outputs of a randomly initialized LLaMA-2-style model are markedly non-uniform and instead exhibit systematic bias: certain dimensions are consistently disfavored, in the sense that they frequently receive the minimum value across random inputs. This extreme bias appears in both logits (top, red) and final hidden representations (bottom, blue). The dashed line denotes the expected frequency under a uniform distribution, and the arrows in the top panel indicate a broken x-axis that omits low-frequency tail ranks. *Upper Right:* During training, models retain weak input-wise correlations in output bias with the base model from which they originate. The resulting correlation distribution is consistently shifted upward relative to correlations computed against independently initialized base models, indicating that initialization-born bias leaves a measurable training-persistent signal. *Lower Right:* The particular dimensions exhibiting strong bias are seed-dependent; the visualization uses the tokens that most frequently receive the minimum logit.

checkpoint and its initialization rank or disfavor the same coordinates across inputs. As a baseline, we compute the corresponding correlation against independently initialized models (seed 1000). As shown in Figure 2 (Upper Right), although the absolute correlations are small, same-initialization correlations are consistently shifted upward relative to the independent baseline and stabilize rather than collapsing to zero. Taken together, these results indicate that initialization leaves a weak yet statistically detectable bias signal that persists throughout training.

4 ALGORITHM

Section 3 shows that models sharing an initialization lineage preserve weak yet consistent agreement on a subset of output dimensions, such as those most strongly disfavored by the untrained model under random inputs. Our objective is therefore to recover, from any two candidate models and without access to the true initialized ancestor, a set of dimensions encoding these bias signals and then to test whether the observed agreement is statistically significant. The complete procedure is summarized in Algorithm 1, with each component detailed below.

Extract identity dimensions between any two models. Let $X = \{x_i\}_{i=1}^n$, where each $x_i \in \mathbb{R}^{\ell \times d}$ denotes a random sequence of length ℓ . Each random input x_i may be instantiated either by uniformly sampling ℓ token IDs from the vocabulary or, alternatively, by directly sampling ℓ independent vectors from a d -dimensional isotropic Gaussian distribution. For any model g , we define its mean response vector as $\bar{g} := \frac{1}{n} \sum_{i=1}^n g(x_i) \in \mathbb{R}^{d_{\text{out}}}$, where d_{out} denotes the output dimensionality. Here, $g(x_i)$ may refer either to the model’s logits or to its final-layer representations. We identify the model’s disfavored dimensions as the bottom- m coordinates of $\bar{g}$:

$$\mathcal{M}_g := \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

We use the mean response vector for bias extraction because it provides a more stable and noise-robust alternative to per-sample argmin frequency, i.e., counting how often each dimension attains the minimum value as illustrated in Figure 2, while also yielding strong empirical performance. A formal comparison is provided in Appendix C.1.

Given two models f and f' , we define their *identity dimensions* as the intersection $\mathcal{S} := \mathcal{M}_f \cap \mathcal{M}_{f'}$. If two models share the same initialization lineage, they should independently recover similar sets

of disfavored dimensions, thereby producing non-trivial overlap between their bottom- m sets. This intersection criterion consequently functions as mutual verification: it suppresses spurious correlations induced by architectural similarity or shared training dynamics, while isolating bias signals more plausibly attributable to shared initialization lineage. When one model is closer to initialization, its disfavored dimensions may also be used directly as an anchor.

Correlation statistics. After restricting both models to the identity dimensions S , we compare their responses across random inputs.¹ For each $j \in S$, we compute a Kendall–Tau rank correlation:

$$\tau_j = \text{KendallTau}(\{f(x_i)_j\}_{i=1}^n, \{f'(x_i)_j\}_{i=1}^n).$$

This produces a collection of correlation statistics $T = \{\tau_j\}_{j \in S}$, capturing the distribution of agreement between the two models over identity dimensions. As observed in Section 3, same-lineage models exhibit distributionally higher agreement on these dimensions than unrelated models.

Hypothesis testing via null distribution. Because the correlation distributions for same-lineage and different-lineage models may overlap, thresholding individual statistics is inadequate. We instead test whether the observed correlations are significantly larger than would be expected under the null hypothesis of no lineage relation. Ideally, this null would be estimated from the correlation distribution between independent models, as in Figure 2. Empirically constructing such a baseline, however, requires comparisons across many independently initialized models and is therefore computationally expensive. As shown in Appendix B, the null correlation distribution between independently initialized models is well approximated by a Gaussian distribution, which motivates a less expensive simulation-based alternative: constructing an empirical null from Gaussian surrogates passed through the same pipeline, as summarized in Algorithm 1.

Although this simulation-based null already offers a strong empirical approximation, it also introduces avoidable randomness and computational overhead: precision is limited by the number of simulation trials, results may vary across runs, and comparisons between empirical distributions typically inherit additional assumptions from the chosen test, such as the one-sided t-test or the Mann–Whitney U test. Because the Kendall–Tau null distribution under weak independence is available in closed form, we further replace the simulation-based procedure with an analytical test, while providing both empirical and analytical implementations in our code repository. Specifically, under the null hypothesis, each Kendall–Tau correlation is approximately zero-mean with known variance σ^2 . By the central limit theorem, as the number of identity dimensions $|S|$ increases, the distribution of $\bar{\tau} = \frac{1}{|S|} \sum_{j \in S} \tau_j$ is

well approximated by $\bar{\tau} \sim \mathcal{N}\left(0, \frac{\sigma^2}{|S|}\right)$. Softmax normalization introduces weak cross-dimensional dependencies, but the use of a relatively high temperature ($T = 10$) mitigates these effects, making the independence assumption a reasonable practical approximation. We empirically validate that this Gaussian approximation closely matches the distribution obtained from the full pipeline (Appendix B). Accordingly, we compute the one-sided z-score $z = \frac{\bar{\tau}}{\sigma/\sqrt{|S|}}$ and its associated p-value $p = 1 - \Phi(z)$,

where Φ denotes the standard normal cumulative distribution function. We reject the null hypothesis and declare shared lineage when $p < 0.01$.

Practical instantiation. Model outputs may be instantiated either as logits or as hidden representations. Although both choices are compatible with our formulation, logits require substantially more random inputs for reliable bias-pattern estimation as vocabulary size grows, thereby imposing considerable computational overhead (see Appendix C.2 for theoretical analysis). For improved efficiency and stability, we therefore use final-layer hidden states as the default model output. The resulting p-value constitutes a direct statistical lineage test rather than a heuristic similarity threshold.

5 EXPERIMENTS

This section is organized into two parts. In Section 5.1, we show that our method behaves as a genuine fingerprint in the Galtonian sense: (i) it enables *birth verification at the seed level*, and (ii) it remains *verifiable across the entire training lifecycle*. Existing baselines, by contrast, fail to support verification in early pretraining and primarily rely on data-dependent signals that break under

¹Before computing the correlations, we apply a row-wise softmax normalization to each sample’s output vector. This removes scale differences across different models and dimensions while preserving the relative ordering that carries identity information for rank-based statistics such as Kendall–Tau. A relatively high temperature is used to avoid degenerate one-hot behavior.

Algorithm 1 Distribution Correlation Test on Identity Dimensions

Require: base model f , suspicious model f' ; random inputs $X = \{x_i\}_{i=1}^n$; fingerprint size m ; significance level α

► **Step 1: Localize biased dimensions**

- 1: Compute average outputs $\bar{f}, \bar{f}'$ over X
- 2: $\mathcal{M}(f), \mathcal{M}(f') \leftarrow$ bottom- m dimensions of $\bar{f}, \bar{f}'$
- 3: $\mathcal{S} \leftarrow \mathcal{M}(f) \cap \mathcal{M}(f')$ (identity dimensions)

► **Step 2: Form correlation distribution**

- 4: Normalize $f(x_i)_{\mathcal{S}}, f'(x_i)_{\mathcal{S}}$ for each $x_i \in X$ across identity dimension
- 5: **for** each $s_j \in \mathcal{S}$ **do**
- 6: $\tau_j \leftarrow \text{KendallTau}(\{f(x_i)_{s_j}\}_{i=1}^n, \{f'(x_i)_{s_j}\}_{i=1}^n)$ (correlation in in input-wise preferences)
- 7: **end for**
- 8: $\mathcal{T} \leftarrow \{\tau_j\}_{j=1}^{|\mathcal{S}|}$

► **Step 3: Hypothesis test against null**

- 9: Construct $\mathcal{T}_{\text{null}}$ by applying the same pipeline to two Gaussian matrices $Y^{(1)}, Y^{(2)} \sim \mathcal{N}(0, I)^{N \times d_{\text{out}}}$
- 10: Test $H_0 : \mathcal{T} = \mathcal{T}_{\text{null}}$ vs. $H_1 : \mathcal{T} > \mathcal{T}_{\text{null}}$
- 11: **Return** SameLineage $\leftarrow \mathbf{1}(p\text{-value} < \alpha)$

Table 1: Comparison of finger- Table 2: Trained models preserve the fingerprint behavior of their print behavior among models ini- initialization (p -value < 0.01). tialized with distinct seeds.

Model Pairs	p -value (> 0.01)	Model Pairs	SeedPrints	Baselines			
			p -value	Intrinsic	REEF	PCS	ICS
s_{42} vs. s_{2000}	0.116	s_{42}^{init} vs. s_{42}^{base}	3.37e-26✓	-0.021×	0.375×	0.580×	0.196×
s_{123} vs. s_{42}	1.000	s_{123}^{init} vs. s_{123}^{base}	3.16e-33✓	0.149×	0.369×	0.581×	0.188×
s_{1000} vs. s_{123}	0.327	s_{1000}^{init} vs. s_{1000}^{base}	1.12e-21✓	-0.252×	0.381×	0.581×	0.188×
s_{2000} vs. s_{1000}	1.000	s_{2000}^{init} vs. s_{2000}^{base}	3.70e-31✓	-0.337×	0.331×	0.581×	0.188×

substantial distributional shifts. We report main results using the analytical null with *hidden state* outputs, and we provide additional empirical-null results using both *logit* and *hidden state* outputs, evaluated with the one-sided t -test and Mann–Whitney U test, in Appendix D.2. In Section 5.3, we assess all methods under practical infringement scenarios using *LeafBench* (Shao et al., 2025). Across 65 distinct model instances spanning 6 representative post-development techniques, our method matches the strongest baselines in post-training settings while remaining robust to heterogeneous deployment transformations.

Baselines We primarily consider four passive fingerprinting baselines based on weights or representations. *Intrinsic fingerprint* (Yoon et al., 2025), also referred to as *PDF* in some papers, compares models using the similarity of layerwise standard-deviation profiles of attention parameters. *REEF* (Zhang et al., 2024) computes centered-kernel-alignment (CKA) similarity between feature representations elicited by the same samples across two models. *PCS* and *ICS* (Zeng et al., 2024), collectively described as *HuRef* in some papers, are weight-similarity methods: PCS flattens all parameters and measures cosine similarity, whereas ICS constructs invariant terms from the weights and measures cosine similarity over those invariants. In Section 5.3, we additionally include the comparatively weaker gradient-based fingerprint *Gradient* (Shao et al., 2025). Following Zhang et al. (2024), we use a **0.8 similarity threshold** for binary decisions.

Throughout the experimental tables, cell colors encode lineage: green marks models from the same source, whereas red marks models from different sources. For example, s_{42}^{init} vs. s_{42}^{base} compares a seed-42 initialized model with its continued-pretrained counterpart and is therefore green. Conversely, s_{2000}^{init} vs. s_{42}^{base} compares a seed-2000 initialization with a model trained from seed 42 and is therefore red. Finally, ✓ indicates a correct decision, while × indicates an error.

5.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. The main paper reports LLaMA-style results, while the corresponding Qwen-style results are deferred to Appendix D; the substantive conclusions are consistent across both architectures.

Table 3: Identical datasets and training orders do not force fingerprint behavior to converge across distinct initializations.

Model Pair	p -value (> 0.01)
s_{42}^{init} vs. s_{123}^{base}	0.573
s_{123}^{init} vs. s_{2000}^{base}	0.724
s_{2000}^{init} vs. s_{1000}^{base}	0.109
s_{1000}^{init} vs. s_{42}^{base}	0.883

Table 4: Persistence of fingerprint evidence under continual training on diverse datasets (base model: seed 1000, corpus openwebtext).

Setting	SeedPrints	Baselines			
Continual corpus (seed)	p -value	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	≈ 0 ✓	1.000✓	0.759×	0.999✓	0.996✓
TinyStories (123)	1.000✓	0.950×	0.658✓	0.332✓	0.012✓
the_stack (1000)	≈ 0 ✓	0.489×	0.557×	0.585×	0.123×
the_stack (123)	1.000✓	0.445✓	0.580✓	0.301✓	0.026✓

Different initialization seeds produce distinct fingerprints. Table 1 reports p -values from correlation tests between model pairs initialized with different random seeds (42, 123, 1000, and 2000). All p -values consistently exceed 0.01, indicating that our method reliably separates models originating from different seeds. Thus, distinct seeds induce distinct fingerprint behavior, enabling models to be distinguished “at birth.”

Training preserves the initialization fingerprint. Table 2 compares each initialization model s_i^{init} with its descendant s^{base} trained on OpenWebText (Gokaslan et al., 2019) (≈ 10 B tokens). Across all seed–model pairs, p -values remain below 0.01, indicating that their bias profiles remain strongly correlated and therefore share a common lineage. In other words, the trained model inherits the fingerprint of its initialization. We also evaluate baseline methods; without exception, they fail to distinguish seeds, suggesting that their apparent separability is produced by training-induced artifacts rather than by initialization itself.

Identical data and order do not make fingerprints converge. In Table 3, all four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ are trained on *exactly the same corpus (OpenWebText) and in the same data order*, with the training seed fixed to lock the curriculum; the only difference is the initialization seed i . Across all cross-seed pairs, p -values remain consistently above 0.01, in sharp contrast to the near-zero values in Table 2. Thus, even under identical data and curriculum, fingerprints remain seed-specific.

Continual training on diverse datasets does not confound the fingerprint. The preceding experiments are intended to demonstrate the distinctive strength of SeedPrints: it can function as a biometric-like fingerprint. From a copyright perspective, the inability of prior methods to separate models with different initialization seeds is not, by itself, a decisive weakness, because initialization seeds have no clear copyright status. The more consequential fragility is that their attribution can be misled by *data distribution, causing them to miss true lineage when the training distribution shifts substantially during continued training*.

In Table 4, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two highly distinct datasets: TinyStories (Eldan & Li, 2023), consisting of synthetic children’s stories, and The Stack (Kocetkov et al., 2022), consisting of permissively licensed GitHub code. We compare (i) true descendants trained from the base against (ii) *distractors* derived from a different base model (seed 123). Because baseline methods are largely insensitive to initialization differences, the distractor base is pretrained on OpenWebText with a different data order before being continually trained on the *same* target corpus to form the distractor. The resulting question is whether an attribution method can identify which descendant truly shares lineage with the base.

We find that all prior baselines fail in the code setting (The Stack), misclassifying true descendants as distractors. This failure indicates that they primarily track domain similarity rather than lineage identity: TinyStories is distributionally closer to the pretraining corpus (OpenWebText), whereas The Stack differs sharply, and this large distributional shift is sufficient to bypass detection. In contrast, our method correctly attributes lineage across both corpora. Thus, the fingerprint recovered by SeedPrints is not merely a proxy for data distribution; it survives substantial domain shift and persists beyond initial pretraining.

5.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. We first ask whether our method can reliably identify a model’s descendants along a known training trajectory. Using a controlled OpenWebText pretraining ex-

Table 5: Fingerprinting results under large-scale finetuning. Each row compares a target model against LLaMA-2-7B. **SeedPrints** reports the p -value from our correlation test, where values below 0.01 indicate strong lineage evidence. The four baselines report similarity scores, for which the decision threshold is 0.8 and larger values indicate stronger evidence.

Model	# Tokens	SeedPrints p	Intrinsic $\uparrow$	REEF ($\uparrow$)	PCS ($\uparrow$)	ICS ($\uparrow$)
Llama-2-finance-7B (Heenan, 2023)	5M	10^{-41955} ✓	1.0000✓	0.9950✓	0.9979✓	0.9952✓
Vicuna-1.5-7B (Chiang et al., 2023)	370M	$10^{-103043}$ ✓	1.0000✓	0.9985✓	0.9985✓	0.9949✓
Wizardmath-7B (Luo et al., 2023)	1.8B	$10^{-180550}$ ✓	1.0000✓	0.9979✓	1.0000✓	0.9994✓
Meditron-7B (Chen et al., 2023)	48B	10^{-42566} ✓	0.9990✓	0.9978✓	1.0000✓	0.9817✓
CodeLlama-7B (Meta AI, 2024)	500B	10^{-3552} ✓	0.9480✓	0.9947✓	0.6863✗	0.3369✗
Llemma-7B (Azerbayev et al., 2023)	700B	10^{-5136} ✓	0.9470✗	0.9984✓	0.6682✗	0.2905✗

periment in which the initialization seed is known, we test lineage at intermediate checkpoints and observe consistent detection across all stages (Appendix D.2.1, Figure 5).

Large-scale pretraining. To evaluate lineage under realistic foundation-model training, we additionally analyze ten checkpoints from the OLMo-2-7B (OLMo et al., 2024) Stage-1 run (5B $\rightarrow$ 3.9T tokens), treating the final checkpoint as the target model and all earlier checkpoints as candidate ancestors. Unlike the controlled OpenWebText setting, this trajectory spans heterogeneous corpora, prolonged optimization, and substantial representation drift. For directional consistency with similarity-based baselines, we plot $1 - p$. All p -values are $\ll 0.01$ and numerically close to zero.

In Figure 1, **all existing baselines fail to verify lineage during early pretraining** (e.g., $ICS \approx 0$, $PCS < 0.3$, and $REEF < 0.6$ within the first **trillion** tokens), whereas SeedPrints remains consistently verifiable and becomes stronger over training. These results expose a critical gap in existing evaluations: **most prior work verifies lineage only after fine-tuning, when lineage detection is substantially easier**. Our findings further show that large-scale pretraining itself can dramatically reshape representations and amplify lineage signals, potentially creating a *false sense of safety*. We therefore argue that lineage verification should prioritize early pretraining, when misuse is hardest to detect, rather than relying exclusively on late-stage checkpoints.

Large-scale finetuning. We further compare our method with existing baselines under standard fine-tuning-stage evaluations. Specifically, we test suspect models fine-tuned from Llama-2-7b (base model) using data volumes ranging from 5 million to 700 billion tokens. The suspect set includes diverse downstream variants such as Llama-2-finance-7b (Heenan, 2023), Vicuna-1.5-7b (Chiang et al., 2023), WizardMath-7b (Luo et al., 2023), Chinese-LLaMA-2-7b (Chen et al., 2023), CodeLlama-7b (Meta AI, 2024), and Llemma-7b (Azerbayev et al., 2023), with fine-tuning data volumes of 5M, 370M, 1.8B, 13B, 500B, and 700B tokens, respectively. As shown in Table 5, our method consistently maintains $p < 0.01$ across all settings.

5.3 ROBUSTNESS UNDER REALISTIC DEPLOYMENTS

Real-world deployments routinely apply *parameter-altering* adaptations to foundation models, including fine-tuning, PEFT, and quantization, all of which can weaken or distort fingerprint signals. Our method is designed to remain effective under precisely these conditions. To evaluate its reliability in such realistic settings, we adopt *LeaFBench* (Shao et al., 2025), a benchmark for language-model copyright auditing.

Across Source Models. We consider two source-model categories for auditing: *Pre-Trained (PT)* and *Instruction-Tuned (IT)*. For each category, the task is to distinguish *derivative* models, obtained through post-training adaptations, from *independent* models trained without the source weights. This setup mirrors common auditing scenarios in which lineage claims must be tested for either a PT base or its IT variant. Our evaluation covers **65** models in total; the complete list of model names and HuggingFace repositories is provided in Appendix E.

Metrics. We evaluate detection performance using (i) the Area Under the ROC Curve (AUC) and (ii) the Kolmogorov–Smirnov (KS) statistic (Berger & Zhou, 2014), which quantifies the maximum separation between the score distributions of derivative and independent models.

Our method produces per-model p -values from statistical tests, which differ fundamentally from the similarity scores used by baseline methods. Whereas similarity scores are linear quantities whose

differences have proportional interpretation, p-values are tail probabilities measuring the rarity of an observation under the null hypothesis and therefore should not be interpreted linearly. To permit comparison with baselines that compute AUC from similarity scores, we convert p-values into scores via $s = 1 - p$. This conversion is not fully aligned with the statistical semantics of p-values and may be suboptimal for our method, because small p-values vary on an exponential scale whereas threshold sweeping is linear, making fine-grained distinctions difficult. Nevertheless, our approach still achieves comparable or superior performance. For evaluation, we (i) compute AUC for all methods by sweeping thresholds over the scores s , and (ii) report the KS statistic as a threshold-free measure of distributional separability. This protocol ensures compatibility with baseline pipelines while retaining the statistical interpretability of our method. We do not report Manhattan distance, because absolute distances between p-values are not meaningful.

In most practical scenarios, calibrating an optimal similarity threshold is infeasible. Although we report AUC for comparability, our method differs fundamentally from prior methods because it provides a direct statistical test. **Given any pair of models, our approach outputs a p-value that enables a definitive decision about whether they share lineage, without relying on tunable thresholds**, thereby supporting reliable verification in practice.

Parameter-Altering Techniques. To assess robustness, we evaluate models produced by six parameter-altering procedures: (1) Instruction tuning (Instruct), (2) General-purpose fine-tuning (Finetune), (3) Parameter-efficient fine-tuning (PEFT), (4) Quantization, (5) Model merging (Merge), and (6) Distillation.

5.3.1 FAMILY-WISE EFFECTIVENESS

There are seven model families in total. For each family, we treat both its pretrained base model (PT) and its instruction-tuned model (IT) as separate source models and compare them against all other models in the 65-model pool. The results are reported in Table 6. We omit the TinyLlama-1.1B families in Table 6, because they are trivial cases for all methods. Each column corresponds to one family, with AUC computed over the corresponding tests using a family-calibrated threshold, whereas the overall score is computed across all test pairs with a global threshold; accordingly, the overall score is not the average of the per-family scores. The three strongest methods—*Intrinsic*, *ICS*, and our *SeedPrints*—are essentially saturated across alterations, with overall AUCs of 0.994, 0.994, and 0.992, respectively, and KS values of 0.989, 0.943, and 0.986. For these methods, family-wise scores are, in most cases, perfect. By contrast, the remaining baselines, *REEF* and *Gradient*, perform substantially worse. These results indicate that, although our method is primarily designed to capture intrinsic seed-level fingerprints, it also functions effectively as a biometric-like fingerprint that supports reliable and persistent identity tracking across diverse model families.

5.3.2 ALTERATION-WISE ROBUSTNESS

Across alteration types, *Instruct* is consistently the easiest setting, with even the weakest baseline, *Gradient*, attaining AUC 0.895, whereas *Finetune* and *Merge* are more challenging overall; for *Finetune*, for example, no method achieves perfect performance. The effects of *PEFT*, *Quantization*, and *Distillation* are more heterogeneous: *REEF* exhibits notable degradation under *Quantization* ($0.996 \rightarrow 0.871$) and *Distillation* ($0.996 \rightarrow 0.858$), while remaining resilient to *PEFT* (0.994); conversely, *PEFT* and *Quantization* are more difficult for *Gradient* ($0.895 \rightarrow 0.776/0.756$). The other three stronger methods remain broadly robust to all transformations, with all scores exceeding 0.9. *SeedPrints* declines slightly to 0.959 on *Merge*, failing on one merge model out of seventeen. Although this represents only a single misclassification, it suggests that weight-space interpolation with unrelated model weights can dilute the initialization signal below the detection threshold. Averaged across all six alteration types, *Intrinsic* still achieves the highest overall AUC, followed closely by *ICS* and *SeedPrints*.

5.4 ABLATION STUDY ON HYPERPARAMETERS

Our method has three hyperparameters: the number of random input sequences n , the length of each random sequence l , and the fingerprint size m . The first two determine the total random-input budget, whereas m specifies how many low-mean coordinates are retained for testing. Throughout the main experiments in Sections 5.2 and 5.3, we set $n=2000$, $l=1024$, and $m=0.1d_{\text{out}}$, where d_{out} denotes hidden-state dimensionality, thereby allowing the method to adapt across open-source model scales. Here, we vary each hyperparameter while fixing the others to their default values (i.e., $n = 2000$,

Table 6: Performance comparison of LLM fingerprinting methods across source models. “PT” and “IT” denote the use of pre-trained and instruction-tuned models as source models, respectively. The per-family breakdown is reported for the 6 families containing both PT and IT variants. The “Overall” scores are computed over all test model pairs rather than averaged across families.

Method	Metric	Qwen2.5-7B		Qwen2.5-14B		Llama3.1-8B		Mistral-7B		Gemma2-2B		Llama2-7B		Overall
		PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	PT	IT	
REEF	AUC $\uparrow$	0.796	0.862	0.947	0.913	1.000	0.999	0.997	0.997	0.963	0.968	0.750	0.706	0.915
	KS Statistic $\uparrow$	0.492	0.633	0.875	0.735	0.987	0.987	0.975	0.980	0.950	0.837	0.750	0.694	0.739
Gradient	AUC $\uparrow$	0.657	0.743	0.763	0.789	0.897	0.904	0.872	0.682	1.000	1.000	0.650	0.768	0.801
	KS Statistic $\uparrow$	0.354	0.423	0.500	0.563	0.646	0.705	0.725	0.500	1.000	1.000	0.275	0.479	0.508
ICS	AUC $\uparrow$	1.000	1.000	0.997	0.997	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic $\uparrow$	1.000	1.000	0.975	0.979	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.943
Intrinsic	AUC $\uparrow$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.994
	KS Statistic $\uparrow$	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.989
SeedPrints	AUC $\uparrow$	0.992	0.994	0.988	0.990	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.992
	KS Statistic $\uparrow$	0.985	0.987	0.975	0.980	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.986

Table 7: Performance of LLM fingerprinting methods across parameter-altering techniques.

Method	Metric	Instruct	Finetune	PEFT	Quantization	Merge	Distillation	Overall Avg
REEF	AUC $\uparrow$	0.996	0.872	0.994	0.871	0.954	0.858	0.924
	KS Statistic $\uparrow$	0.949	0.637	0.940	0.812	0.851	0.615	0.801
Gradient	AUC $\uparrow$	0.895	0.729	0.776	0.756	0.832	0.887	0.812
	KS Statistic $\uparrow$	0.756	0.441	0.469	0.586	0.519	0.679	0.575
ICS	AUC $\uparrow$	1.000	0.987	1.000	1.000	0.981	1.000	0.995
	KS Statistic $\uparrow$	1.000	0.925	1.000	1.000	0.941	1.000	0.978
Intrinsic	AUC $\uparrow$	1.000	0.980	1.000	1.000	1.000	1.000	0.997
	KS Statistic $\uparrow$	1.000	0.971	1.000	1.000	1.000	1.000	0.995
SeedPrints	AUC $\uparrow$	1.000	0.995	0.990	1.000	0.959	1.000	0.991
	KS Statistic $\uparrow$	1.000	0.990	0.980	1.000	0.941	1.000	0.985

$m = 400$, $l = 768$), and we report empirical accuracy and false-positive rates under the fixed significance level $\alpha = 0.01$ in Table 8.

Increasing n and l consistently improves identification performance. This pattern is explained by the theoretical analysis in Section C.2, which shows that accurate recovery of the underlying population-level bias requires the number of queries to satisfy $n = \Omega\left(\frac{\log d_{\text{out}}}{\gamma^2}\right)$, where γ is the bias margin separating biased coordinates from the remaining dimensions. Thus, each target output dimension requires a minimum number of samples for accurate detection, explaining the suboptimality of small n . Conversely, because γ increases with l through accumulation effects, longer sequences strengthen separation. Under a fixed and insufficient number of sequences, increasing l therefore reduces the effective query requirement and improves detection.

The effect of m is non-monotonic. When m is too small, the selected coordinate set is unstable because too few empirical observations are available to form a reliable distribution. When m is too large, many irrelevant coordinates are included, introducing noise that shrinks γ and would require quadratically more samples to compensate. Empirically, selecting m in the range 200–400 performs well across model sizes, aligning with our heuristic $m \approx 0.1d_{\text{out}}$ for typical $d_{\text{out}} \in [3000, 5000]$.

6 CONCLUSION

In this work, we introduced *SeedPrints*, a stronger and more intrinsic conception of LLM fingerprinting that traces model lineage back to random initialization. We showed that initialization itself leaves a persistent “Galtonian” fingerprint on neural language models: even before training, models already display stable, seed-dependent token-preference patterns, and these biases remain

Table 8: Hyperparameter ablation under significance level $\alpha = 0.01$.

Ablation on n	200	500	2000
ACC	0.7368	0.7778	0.9375
Empirical FPR	0	0	0
Ablation on l	512	768	1024
ACC	0.9048	0.9375	1.0000
Empirical FPR	0.04	0	0
Ablation on m	200	400	800
ACC	0.9375	0.9375	0.875
Empirical FPR	0.04	0	0.1463

statistically and measurably detectable throughout optimization. Building on this observation, we proposed a distribution-correlation test over *identity dimensions* that detects shared lineage with calibrated significance. Across LLaMA- and Qwen-style families, as well as a broad suite of realistic deployment transformations—including continued pretraining on divergent corpora, instruction tuning, PEFT, quantization, merging, and distillation—SeedPrints enables “birth-to-lifecycle” verification and remains robust under domain shift, thereby offering a practical instrument for provenance and copyright auditing.

Limitations and Future Directions This work seeks to uncover fingerprint signals rooted in initialization-stage bias. The proposed method requires access to model outputs, such as logits or hidden states, and may not apply when only restricted API access is available. Nevertheless, we emphasize that the search for *true fingerprints* for LLMs remains in an early stage: most existing approaches, especially black-box methods, depend on signals that emerge only in later training stages or on alignment-induced artifacts, and therefore cannot robustly protect models at the pretrained stage. Reliable all-stage fingerprints are consequently increasingly urgent. See full discussions in Section A.2.

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788
789
790
791
792
793
794
795
796
797
798
799
800
801
802
803
804
805
806
807
808
809

Appendix Contents

A	LLM Usage and Limitation Discussions	16
A.1	LLM Usage	16
A.2	Limitations and Future Directions	16
B	Validation of the Empirical and Analytical Null Distribution	16
B.1	Empirical True Null from Independently Initialized Models	16
B.2	Gaussian Surrogate as a Simulation-Based Approximation	16
B.3	Analytical Approximation under Weak Dependence	17
C	Theoretical Properties of Our Methods	18
C.1	Stability Analysis of Mean-Based Identity Extraction vs. Per-Sample Argmin Voting	19
C.1.1	Stability of Sample-Mean-Based Extraction	19
C.1.2	Instability of Per-Sample Argmin Voting	20
C.2	Influence of Hyperparameters	22
D	More Experiment Details	23
D.1	Implementation Details and Licenses	23
D.2	Complementary SeedPrints Results using simulation-based null	23
D.2.1	Birth-to-Lifecycle “Biometric” Fingerprinting	23
D.2.2	All-stage verifiable fingerprints	25
D.3	Cross-Size Fingerprint Within the Qwen-2.5 series	26
E	Model Catalog and Decoding Settings	26
E.1	Model list by family	27
E.1.1	Qwen-2.5-7B	27
E.1.2	Qwen2.5-14B	27
E.1.3	Llama-3.1-8B	27
E.1.4	Mistral-7B-v0.3	28
E.1.5	Gemma-2-2B	28
E.1.6	Llama-2-7B	28
E.1.7	TinyLlama-1.1B	28
E.2	Sample Results	29

A LLM USAGE AND LIMITATION DISCUSSIONS

A.1 LLM USAGE

We use AI assistants, including ChatGPT and Gemini, for writing and formatting support, primarily for grammar and style checking, caption clarification, and other surface-level editorial refinements. For programming-related tasks, we occasionally use GitHub Copilot and Claude as coding assistants, for example to obtain code-completion suggestions and debugging hints.

A.2 LIMITATIONS AND FUTURE DIRECTIONS

Although this work identifies seed-born, training-persistent biases that can serve as fingerprint signals, the proposed method remains within the white-box paradigm. It may therefore be inapplicable when only restricted API access is available, for example when responses are deterministic, safety-filtered, or otherwise deprived of the required output information. We nevertheless emphasize that the search for *true fingerprints* for LLMs remains at a very early stage, even in white-box settings. Most existing fingerprinting approaches depend on training-induced signatures or alignment behaviors and consequently cannot robustly protect pretrained or early-stage models. By analogy to human fingerprints, which remain stable despite changes in age or environment, we argue that a genuine LLM fingerprint should persist across continued training and deployment contexts rather than merely reflect incidental training artifacts. Achieving this goal—particularly under black-box access, where a substantial performance gap relative to white-box methods persists—remains a difficult open direction requiring considerable future work. We hope this study motivates further research on principled and robust fingerprinting frameworks for large models.

B VALIDATION OF THE EMPIRICAL AND ANALYTICAL NULL DISTRIBUTION

The hypothesis test in Section 4 determines whether two models share lineage by comparing the distribution of their correlation statistics over random inputs with an appropriate null distribution. This null distribution represents the correlation statistics expected between two *independent models evaluated on random inputs*.

We consider two approximations to this null distribution: (i) an empirical simulation-based null obtained by applying the full pipeline to Gaussian surrogate outputs, and (ii) an analytical null derived from the known distribution of Kendall–Tau correlations. This section justifies these approximations in three stages: first, by characterizing the true null using independently initialized models; second, by introducing the Gaussian surrogate as a tractable simulation-based proxy; and third, by validating the analytical approximation.

B.1 EMPIRICAL TRUE NULL FROM INDEPENDENTLY INITIALIZED MODELS

The ideal null distribution corresponds to the correlation statistics obtained from two independent models evaluated on random inputs. The cleanest instantiation of this independence is a comparison between two independently initialized models, with different random seeds and no training. We estimate this empirical null by evaluating ~ 2500 such pairs and computing Kendall–Tau correlations on 10,000 random inputs, following Section 3. Figure 3 shows that the resulting true null is empirically well described by a zero-centered Gaussian distribution.

B.2 GAUSSIAN SURROGATE AS A SIMULATION-BASED APPROXIMATION

Direct estimation of the null distribution from independently initialized models is computationally expensive, because it requires access to many models and repeated evaluations. To obtain a tractable alternative, we introduce a simulation-based null that uses Gaussian surrogates as proxies for model outputs under the null.

This choice is motivated by two considerations. First, Gaussian approximation of neural-network outputs is widely used in the auditing literature and is supported by both empirical and theoretical studies (Carlini et al., 2022; Lee et al., 2019). Second, our pipeline depends only on relative ordering and selection operations, such as rank correlations and bottom- m selection, and does not exploit

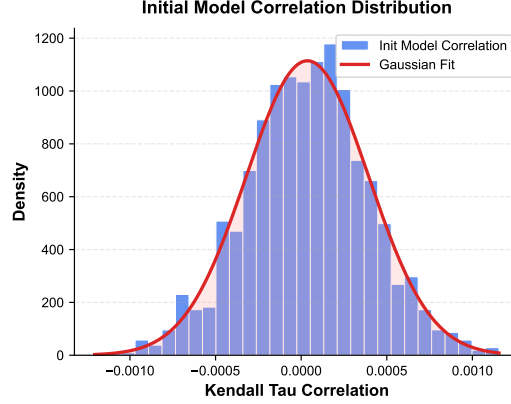


Figure 3: The empirical correlation distribution over ~ 2500 independently initialized model pairs, evaluated on random inputs, closely matches a zero-centered Gaussian distribution.

semantic structure in the inputs. Moreover, because randomly initialized models possess independent parameter matrices, their null outputs should not exhibit strong structured dependencies. Sampling independent Gaussian matrices therefore provides a simple, tractable proxy for the null behavior of independently initialized models.

We construct the surrogate null by sampling Gaussian matrices and applying the full pipeline, including identity-dimension selection and correlation computation:

1. Sample two Gaussian matrices of shape $\mathbb{R}^{N \times d_{\text{out}}}$,
2. Extract the most biased dimensions by selecting the bottom- m ranked average dimensions,
3. Take the intersection set $\mathcal{S}$ and compute correlations on it.

This simulation-based construction applies the same selection procedure to Gaussian samples and thereby captures dependency patterns introduced by the pipeline itself, including those induced by selecting $\mathcal{S}$ across dimensions. As shown in Figure 4 (middle), the resulting distribution matches the empirical null from independently initialized models in its centered Gaussian shape, indicating that the Gaussian surrogate is an effective simulation-based approximation to the true null.

B.3 ANALYTICAL APPROXIMATION UNDER WEAK DEPENDENCE

Although the Gaussian surrogate captures the behavior of the null distribution under the full pipeline, it still requires repeated sampling and introduces estimation randomness. We therefore also propose a more efficient analytical approximation.

Under the null hypothesis that the two models are independent², each Kendall–Tau correlation τ_j is approximately zero-mean with known variance σ^2 . If correlations across identity dimensions were independent, the central limit theorem would imply

$$\bar{\tau} = \frac{1}{|\mathcal{S}|} \sum_{j \in \mathcal{S}} \tau_j \sim \mathcal{N}\left(0, \frac{\sigma^2}{|\mathcal{S}|}\right).$$

In practice, softmax normalization induces dependencies across dimensions. However, Figure 4 shows that the analytical Gaussian closely matches the distribution produced by the Gaussian-surrogate pipeline across settings with varying numbers of random inputs n and output dimensions D . This agreement suggests that the dependencies are sufficiently weak and do not materially compromise the approximation. A plausible explanation is that the relatively high temperature ($T = 10$) smooths the output distribution and reduces inter-dimensional coupling.

Together, these results justify the analytical Gaussian null used in Section 4 as an efficient, deterministic, and accurate approximation, eliminating the need for costly empirical baseline construction.

²As discussed in Section B.2, although outputs may exhibit weak dependencies due to shared inputs, this approximation remains accurate empirically.

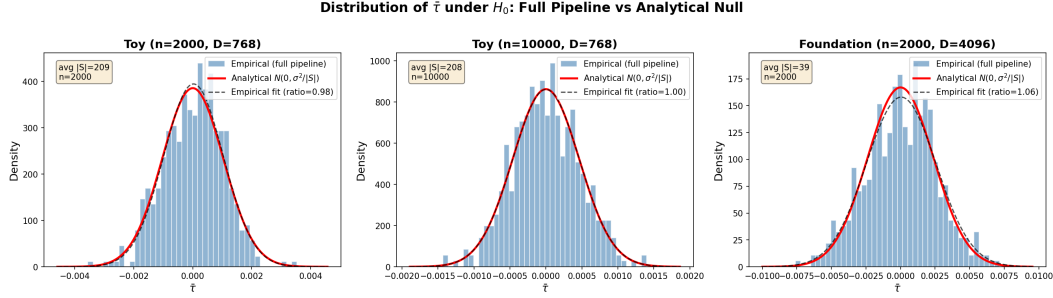


Figure 4: Validation of the analytical Gaussian null. We apply the full pipeline, including identity-dimension selection and correlation aggregation, to Gaussian surrogate outputs and compare the resulting distribution of $\bar{\tau}$ with the analytical Gaussian $\mathcal{N}(0, \sigma^2/|S|)$. The close agreement indicates that the analytical approximation accurately captures null behavior despite weak dependencies introduced by the pipeline.

On variance scaling. The true null obtained from randomly initialized models has slightly smaller variance than the Gaussian null approximation. This mismatch does not affect our hypothesis-test results. The primary simulation-based-null evaluation in Section D.2 relies on the Mann-Whitney U-test, which is rank-based and therefore invariant to positive scaling: for any $c > 0$, $x_i < x_j \Leftrightarrow cx_i < cx_j$. Accordingly, all rank statistics and rejection thresholds remain unchanged. For tests that depend on absolute variance, we additionally evaluate variance-sensitive statistics such as the t -test, whose statistic scales as $t' = t/c$ when variance decreases. Because $c < 1$ in our setting, scaling reduces the magnitude of t and makes the test more conservative rather than artificially inflating significance. Empirically, across the extensive experiments in Section D.2, variance scaling does not affect the power of the t -test, indicating that the distributional separation between the alternative hypothesis and the null is substantially larger than the scaled variance.

For the analytical null used in the main paper, variance is explicitly incorporated into the normalization of the z -score, so any scale discrepancy is absorbed by the denominator. Because the analytical variance is slightly larger than the empirical variance, the resulting z -scores are smaller, making the test conservative and guarding against false positives.

C THEORETICAL PROPERTIES OF OUR METHODS

Our goal is to identify the output dimensions that an initialized model is least likely to predict under random inputs. These dimensions correspond to directions inherently disfavored by the model because of initialization-induced bias, rather than because of the input distribution, and they therefore form the basis of a stable fingerprint signal.

Setup. Consider a model g with fixed parameters and random inputs x_i (uniform random tokens or random embeddings as in the main paper). For each output dimension $j \in [d_{\text{out}}]$, $g_j(x)$ is thus a random variable and its population mean is $\bar{g}_j := \mathbb{E}[g_j(x)]$. The population-level directions we aim to capture correspond to the m output dimensions with the smallest expected values, formally defined as

$$\mathcal{M} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \bar{g}_j.$$

Let $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ denote the sorted means in non-decreasing order, and write

$$\Gamma(m) := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$$

for the population “gap” between the m -th and $(m+1)$ -th smallest means. This gap measures the separation between the target set $\mathcal{M}$ and the remaining coordinates. Importantly, for a fixed m and input distribution, $\Gamma(m)$ depends solely on the model’s inherent initialization bias. *Increasing the length of random input sequences, however, amplifies this bias through concentration, making the population-level separation more pronounced.*

Two natural empirical strategies can be used to extract these population-level least-likely dimensions:

1. **Sample-mean-based extraction (ours).** Estimate the expectation of each output coordinate by the empirical mean over random inputs $X = \{x_i\}_{i=1}^n$ and select the m coordinates with the smallest value. Following the notation in Section 4, where g denotes the model (either the base model f or the suspect model f'):

$$\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i), \quad \widehat{\mathcal{M}}^{\text{mean}} = \arg \min_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} \hat{g}_j.$$

2. **Per-sample argmin voting.** For each random input $x_i \in \mathbb{R}^{\ell \times d}$, select the smallest-value coordinate and then take the m most frequent dimensions:

$$j^*(x_i) = \arg \min_{j \in \{1, \dots, d_{\text{out}}\}} g_j(x_i), \quad \widehat{\mathcal{M}}^{\text{vote}} = \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{i=1}^n \mathbf{1}\{j^*(x_i) \in J\}.$$

Let $p_j := \Pr(j^*(x) = j)$ denote the population probability that index j attains the minimum, and let

$$\mathcal{M}_{\text{vote}} := \arg \max_{J \subseteq \{1, \dots, d_{\text{out}}\}, |J|=m} \sum_{j \in J} p_j$$

be the population top- m set under voting.

Both approaches are intuitively reasonable. As shown below, however, the sample-mean-based strategy is provably stable, whereas per-sample argmin voting is discontinuous and can flip under arbitrarily small perturbations, making it intrinsically noise-sensitive.

C.1 STABILITY ANALYSIS OF MEAN-BASED IDENTITY EXTRACTION VS. PER-SAMPLE ARGMIN VOTING

C.1.1 STABILITY OF SAMPLE-MEAN-BASED EXTRACTION

We first show that the sample-mean-based operator, which selects the m smallest coordinates of the empirical mean vector, is stable under small perturbations to model outputs. These perturbations represent realistic output variations from the *same* model, such as numerical noise or evaluation under different precision modes.

Formally, we model perturbed outputs as $g'(x) = g(x) + \eta(x)$, where $\eta(x) \in \mathbb{R}^{d_{\text{out}}}$ represents stochastic deviations. We assume that for each sample x_i and coordinate j , the random variables $\eta_j(x_i)$ are independent across samples (not necessarily across coordinates) and σ -sub-Gaussian with $\mathbb{E}[\eta_j(x_i)] = 0$.³

Lemma C.1 (Stability of sample-mean-based extraction). *Let $\hat{g}$ and $\hat{g}'$ be the empirical means over $X = \{x_i\}_{i=1}^n$, i.e., $\hat{g} = \frac{1}{n} \sum_{i=1}^n g(x_i)$ and $\hat{g}' = \frac{1}{n} \sum_{i=1}^n g'(x_i) = \hat{g} + \frac{1}{n} \sum_{i=1}^n \eta(x_i)$. Let $\widehat{\Gamma}(m)$ denote the empirical gap at rank m . Then,*

$$\Pr(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

Interpretation. The lemma gives a quantitative upper bound on the probability that the sample-mean-based extractor changes its selected set under perturbations. This instability probability decreases exponentially in the number of averaged samples n , because empirical averaging reduces the variance proxy from σ^2 per sample to σ^2/n .

Proof. We condition on the sample $X = \{x_i\}_{i=1}^n$ throughout the proof, so that $\hat{g}$ and $\widehat{\Gamma}(m)$ are fixed, and all probabilities are taken over the randomness of $\{\eta(x_i)\}_{i=1}^n$ only.

³This sub-Gaussian assumption is mild: bounded random variables are σ -sub-Gaussian (Hoeffding's lemma), and common initialization schemes together with bounded activations yield light-tailed behavior in both initialized and trained networks. The Gaussian case $\eta_j(x_i) \sim \mathcal{N}(0, \sigma^2)$ is a special case of this assumption.

For any coordinate j , by definition of $\hat{g}'_j$, $\hat{g}'_j - \hat{g}_j = \frac{1}{n} \sum_{i=1}^n \eta_j(x_i)$. Each $\eta_j(x_i)$ is mean-zero and σ -sub-Gaussian, so their average is mean-zero and $(\sigma/\sqrt{n})$ -sub-Gaussian: the variance proxy scales as $1/n$ under averaging. Hence, for any $\varepsilon > 0$, by the standard sub-Gaussian tail bound,

$$\Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2 \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (1)$$

Now consider the event that *any* coordinate fluctuates by more than ε :

$$\left\{ \max_{1 \leq j \leq d_{\text{out}}} |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \right\}.$$

Taking a union bound over all d_{out} coordinates yields

$$\Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}'_j - \hat{g}_j| \geq \varepsilon \mid X) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right). \quad (2)$$

We now show that

$$\left\{ \widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g) \right\} \subseteq \left\{ \max_j |\hat{g}'_j - \hat{g}_j| \geq \widehat{\Gamma}(m)/2 \right\}.$$

Equivalently, whenever $\max_j |\hat{g}'_j - \hat{g}_j| < \widehat{\Gamma}(m)/2$, the bottom- m identity set remains unchanged.

Let $j_{(1)}, \dots, j_{(d_{\text{out}})}$ be a permutation of $\{1, \dots, d_{\text{out}}\}$ such that $\hat{g}_{j_{(1)}} \leq \hat{g}_{j_{(2)}} \leq \dots \leq \hat{g}_{j_{(d_{\text{out}})}}$. Recall that the empirical gap at rank m is $\widehat{\Gamma}(m) := \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} > 0$. Denote $\widehat{\mathcal{M}}^{\text{mean}}(g) := \{j_{(1)}, \dots, j_{(m)}\}$ and $\widehat{\mathcal{M}}_c^{\text{mean}}(g) := \{j_{(m+1)}, \dots, j_{(d_{\text{out}})}\}$. For any $i \in \widehat{\mathcal{M}}^{\text{mean}}(g)$ and any $\ell \in \widehat{\mathcal{M}}_c^{\text{mean}}(g)$, $\hat{g}_\ell - \hat{g}_i \geq \hat{g}_{j_{(m+1)}} - \hat{g}_{j_{(m)}} = \widehat{\Gamma}(m)$. With $|\hat{g}'_i - \hat{g}_i| < \widehat{\Gamma}(m)/2$ and $|\hat{g}'_\ell - \hat{g}_\ell| < \widehat{\Gamma}(m)/2$, we have

$$\hat{g}'_\ell - \hat{g}'_i \geq (\hat{g}_\ell - \frac{\widehat{\Gamma}(m)}{2}) - (\hat{g}_i + \frac{\widehat{\Gamma}(m)}{2}) \geq \widehat{\Gamma}(m) - \widehat{\Gamma}(m) = 0.$$

Consequently, under this deviation bound, no coordinate in $\widehat{\mathcal{M}}_c^{\text{mean}}(g)$ can become smaller than any coordinate in $\widehat{\mathcal{M}}^{\text{mean}}(g)$, and therefore

$$\widehat{\mathcal{M}}^{\text{mean}}(g') = \widehat{\mathcal{M}}^{\text{mean}}(g).$$

Because $\widehat{\Gamma}(m)$ is fixed after conditioning on X , substituting $\varepsilon = \widehat{\Gamma}(m)/2$ into (2) yields

$$\Pr\left(\widehat{\mathcal{M}}^{\text{mean}}(g') \neq \widehat{\mathcal{M}}^{\text{mean}}(g)\right) \leq \Pr\left(\max_j |\hat{g}'_j - \hat{g}_j| \geq \widehat{\Gamma}(m)/2 \mid X\right) \leq 2d_{\text{out}} \exp\left(-\frac{n\widehat{\Gamma}(m)^2}{8\sigma^2}\right).$$

□

C.1.2 INSTABILITY OF PER-SAMPLE ARGMIN VOTING

In contrast, per-sample voting applies a discontinuous winner-take-all operator to each input, so arbitrarily small output perturbations can change decisions with non-vanishing probability. We continue to use the same stochastic perturbation model $g'(x) = g(x) + \eta(x)$.

Lemma C.2 (Instability of per-sample argmin voting). *For a fixed input x , let*

$$j^*(x; g) := \arg \min_j g_j(x), \quad \Delta_g(x) := \min_{j \neq j^*(x; g)} \{g_j(x) - g_{j^*(x; g)}(x)\} > 0$$

denote the argmin and its margin. Then, for any fixed x ,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right),$$

and for n i.i.d. inputs $x_1, \dots, x_n$,

$$\Pr\left(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)\right) \geq 1 - (1 - \alpha)^n,$$

where $\alpha := \mathbb{E}_x \left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) \right]$.

Interpretation. Unlike mean extraction, voting does not benefit from variance reduction: each input induces a direct hard decision based on noisy outputs, so increasing n increases the number of opportunities for index flips.

Proof. We first bound the per-input change probability $\Pr(j^*(x; g') \neq j^*(x; g) \mid x)$. Fix x and write $j^* := j^*(x; g)$ for brevity. By definition of the margin, $\Delta_g(x) = \min_{j \neq j^*} \{g_j(x) - g_{j^*}(x)\}$. Let $j^{(2)}$ be an index attaining this minimum, i.e., $g_{j^{(2)}}(x) - g_{j^*}(x) = \Delta_g(x)$.

Under the perturbation $g'(x) = g(x) + \eta(x)$, we have

$$g'_{j^{(2)}}(x) - g'_{j^*}(x) = [g_{j^{(2)}}(x) - g_{j^*}(x)] + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] = \Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)].$$

Since $\eta_{j^{(2)}}(x)$ and $\eta_{j^*}(x)$ are independent $\mathcal{N}(0, \sigma^2)$ variables, their difference is Gaussian with

$$\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \sim \mathcal{N}(0, 2\sigma^2).$$

The event that the argmin changes, $j^*(x; g') \neq j^*(x; g)$, certainly occurs whenever $g'_{j^{(2)}}(x) \leq g'_{j^*}(x)$. Hence,

$$\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x).$$

Using the expression above,

$$\Pr(g'_{j^{(2)}}(x) \leq g'_{j^*}(x) \mid x) = \Pr(g'_{j^{(2)}}(x) - g'_{j^*}(x) \leq 0 \mid x) = \Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x).$$

Let $Z \sim \mathcal{N}(0, 1)$. Since $\eta_{j^{(2)}}(x) - \eta_{j^*}(x) \stackrel{d}{=} \sqrt{2}\sigma Z$, we obtain

$$\Pr(\Delta_g(x) + [\eta_{j^{(2)}}(x) - \eta_{j^*}(x)] \leq 0 \mid x) = \Pr\left(Z \leq -\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right) = \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right).$$

Therefore, $\Pr(j^*(x; g') \neq j^*(x; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)$.

Now consider n i.i.d. inputs $x_1, \dots, x_n$. Conditional on the inputs, the events

$$E_i := \{j^*(x_i; g') \neq j^*(x_i; g)\} \quad E_i^c := \{j^*(x_i; g') = j^*(x_i; g)\}$$

are independent across i , because the perturbations $\eta(x_i)$ are independent across i . Moreover,

$$\Pr(E_i \mid x_i) = \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right) \geq \Phi\left(-\frac{\Delta_g(x_i)}{\sqrt{2}\sigma}\right).$$

Define $q(x_i) := \Pr(j^*(x_i; g') \neq j^*(x_i; g) \mid x)$, and $\alpha := \mathbb{E}_x\left[\Phi\left(-\frac{\Delta_g(x)}{\sqrt{2}\sigma}\right)\right]$. Taking expectations over the randomness of x_i , we obtain $\mathbb{E}[\Pr(E_i \mid x_i)] = \alpha$. Therefore, by independence across i ,

$$\Pr(E_1^c \cap \dots \cap E_n^c) = \mathbb{E}\left[\prod_{i=1}^n (1 - q(x_i))\right] \leq \mathbb{E}\left[\prod_{i=1}^n (1 - \Phi(-\Delta_g(x_i)/(\sqrt{2}\sigma)))\right] \leq (1 - \alpha)^n,$$

where the last inequality uses i.i.d. inputs and Jensen's inequality applied to the convex function $u \mapsto \log(1 - u)$ on $u \in [0, 1)$. Therefore,

$$\Pr(\exists i \leq n : j^*(x_i; g') \neq j^*(x_i; g)) = 1 - \Pr(E_1^c \cap \dots \cap E_n^c) \geq 1 - (1 - \alpha)^n.$$

Finally, the empirical voting set $\widehat{\mathcal{M}}^{\text{vote}}$ is determined by the empirical counts of dimensions $\{j^*(x_i; g)\}_{i=1}^n$. If the empirical gap between the m -th and $(m+1)$ -th most frequent dimensions is of order $O(1/n)$, corresponding to only a constant number of votes, then changing $j^*(x_i)$ for a single input x_i can swap the ordering across this boundary and alter the top- m set. Combining this observation with the lower bound $1 - (1 - \alpha)^n$ shows that, in such regimes, the probability that $\widehat{\mathcal{M}}^{\text{vote}}(g') \neq \widehat{\mathcal{M}}^{\text{vote}}(g)$ is bounded away from zero uniformly in n , i.e., per-sample argmin voting does not enjoy variance reduction with more samples. $\square$

C.2 INFLUENCE OF HYPERPARAMETERS

We now analyze the number of random inputs required for the empirical mean estimator to recover the true bias. We reuse the notation from **Setup**: each output coordinate $g_j(x)$ is a random variable under random inputs $x \sim \mathcal{D}$, with population mean $\bar{g}_j := \mathbb{E}[g_j(x)]$. $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{\text{out}})})$ denotes the sorted means in non-decreasing order, and $\gamma := \bar{g}_{(m+1)} - \bar{g}_{(m)} > 0$ denotes the separation margin. We estimate $\bar{g}$ using n i.i.d. random inputs:

$$\hat{g}_j := \frac{1}{n} \sum_{i=1}^n g_j(x_i), \quad \hat{g} = (\hat{g}_1, \dots, \hat{g}_{d_{\text{out}}}).$$

The target dimension set is $\mathcal{M}$ with size m , and its empirical estimate is denoted by $\widehat{\mathcal{M}}$.

Lemma C.3 (Sufficient condition for recovering the true set). *If $\|\hat{g} - \bar{g}\|_\infty \leq \varepsilon$ and $2\varepsilon < \gamma$, then the empirical bottom- m set equals the true set: $\widehat{\mathcal{M}} = \mathcal{M}$.*

Proof. For any $j \in \mathcal{M}$ and $j' \notin \mathcal{M}$, we have $\bar{g}_j \leq \bar{g}_{(m)}$ and $\bar{g}_{j'} \geq \bar{g}_{(m+1)}$. The ℓ_∞ bound gives

$$\hat{g}_j \leq \bar{g}_{(m)} + \varepsilon, \quad \hat{g}_{j'} \geq \bar{g}_{(m+1)} - \varepsilon.$$

Thus,

$$\hat{g}_j - \hat{g}_{j'} \leq -(\gamma - 2\varepsilon) < 0,$$

so no outside index can enter, and no inside index can leave, the bottom- m set. $\square$

We next interpret this recovery condition in terms of hyperparameters. We again adopt the sub-Gaussian assumption used in Section C.1. This assumption is mild: by Hoeffding's inequality, any bounded random variable is σ -sub-Gaussian with σ proportional to its range. In our setting, each $g_j(x)$ corresponds to a logit or final hidden activation, which is typically bounded or light-tailed in both initialized networks, due to controlled parameter initialization, and trained networks.

Corollary C.4 (Sample complexity under sub-Gaussian coordinates). *Suppose each $g_j(x)$ is σ -sub-Gaussian. Then with probability at least $1 - \delta$,*

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Consequently, if

$$n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right),$$

then $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$.

Proof. For fixed j , sub-Gaussian concentration implies

$$\Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2 \exp(-n\varepsilon^2/2\sigma^2).$$

A union bound over all d_{out} coordinates yields

$$\Pr\left(\|\hat{g} - \bar{g}\|_\infty > \varepsilon\right) \leq \sum_{j=1}^{d_{\text{out}}} \Pr(|\hat{g}_j - \bar{g}_j| > \varepsilon) \leq 2d_{\text{out}} \exp\left(-\frac{n\varepsilon^2}{2\sigma^2}\right).$$

Set the right-hand side to δ and solve for ε to obtain, with probability at least $1 - \delta$,

$$\|\hat{g} - \bar{g}\|_\infty \leq \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

Now choose

$$\varepsilon := \sigma \sqrt{\frac{2 \log(2d_{\text{out}}/\delta)}{n}}.$$

If $n \geq \frac{8\sigma^2}{\gamma^2} \log\left(\frac{2d_{\text{out}}}{\delta}\right)$ (i.e., if $2\varepsilon < \gamma$), then Lemma C.3 applies and implies $\widehat{\mathcal{M}} = \mathcal{M}$ with probability at least $1 - \delta$. $\square$

Table 9: Fingerprint behavior induced by distinct random initializations across model families.

(a) LLaMA-style model results					(b) Qwen-style model results				
Seed Pair	Logits Output		Hidden State		Seed Pair	Logits Output		Hidden State	
	<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test		<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test
s_{42} vs. s_{2000}	0.404	0.456	0.357	0.532	s_{123} vs. s_{1000}	0.741	0.727	0.094	0.074
s_{123} vs. s_{42}	0.214	0.295	0.678	0.565	s_{1000} vs. s_{123}	0.954	0.971	0.125	0.094
s_{1000} vs. s_{123}	0.219	0.246	0.363	0.335	s_{42} vs. s_{2000}	0.273	0.360	0.451	0.529
s_{2000} vs. s_{1000}	0.282	0.291	0.434	0.481	s_{2000} vs. s_{42}	0.215	0.206	0.230	0.295

Table 10: (LLaMA-style models) Trained models preserve the fingerprint behavior of their initialization (p -value < 0.01).

Model Pair	Logits Output		Hidden State		Baselines			
	<i>t</i> -test	<i>u</i> -test	<i>t</i> -test	<i>u</i> -test	Intrinsic	REEF	PCS	ICS
s_{42}^{init} vs. s_{42}^{base}	3.33e-3✓	1.02e-3✓	2.20e-8✓	6.28e-8✓	-0.021×	0.375×	0.580×	0.196×
s_{123}^{init} vs. s_{123}^{base}	2.06e-3✓	7.33e-3✓	7.09e-6✓	1.37e-5✓	0.149×	0.369×	0.581×	0.188×
s_{1000}^{init} vs. s_{1000}^{base}	2.44e-3✓	4.14e-3✓	5.58e-4✓	2.81e-3✓	-0.252×	0.381×	0.581×	0.188×
s_{2000}^{init} vs. s_{2000}^{base}	5.63e-3✓	6.76e-3✓	4.00e-10✓	1.27e-9✓	-0.337×	0.331×	0.581×	0.188×

Therefore, accurate recovery requires $n = \Omega\left(\frac{\log d_{out}}{\gamma^2}\right)$: the number of random queries grows logarithmically with output dimensionality but inversely with the squared separation margin γ . The margin γ is shaped by two factors:

- **Intrinsic bias strength.** Longer random input sequences amplify initialization-induced bias via concentration, increasing the population separation between coordinates.
- **Choice of m .** Different choices of m correspond to different parts of the ranked spectrum $(\bar{g}_{(1)}, \dots, \bar{g}_{(d_{out})})$, thus yielding different effective margins in practice.

The former is controlled by input length l , whereas the latter is selected empirically from the observed gaps in the mean spectrum.

D MORE EXPERIMENT DETAILS

D.1 IMPLEMENTATION DETAILS AND LICENSES

We train all models using the Hugging Face Transformers Trainer (Wolf et al., 2020), with Accelerate (Gugger et al., 2022) used for distributed runs. All open-source models are loaded from their official Hugging Face releases and used under their original licenses: Llama models under the Meta Llama Community License and other models under Apache-2.0. All datasets are downloaded through the Hugging Face Datasets library, which is Apache-2.0 licensed; dataset contents follow their respective stated licenses.

D.2 COMPLEMENTARY SEEDPRINTS RESULTS USING SIMULATION-BASED NULL

This section contains the complementary results for Section 5. We report results using the empirical simulation-based null in Algorithm 1 with both *logits* and *hidden state* outputs, and we test with both the one-sided *t*-test (*t*-test) and Mann–Whitney *U* test (*U* test).

D.2.1 BIRTH-TO-LIFECYCLE “BIOMETRIC” FINGERPRINTING

We train 12-layer, 12-head LLaMA-style models (Touvron et al., 2023) with RoPE (Su et al., 2021) and Qwen-style models (Team, 2024) from scratch. Because the simulation-based random baseline is stochastic, we report p -values averaged over 10 independent trials and adopt a significance level of $\alpha = 0.01$. Importantly, the absolute magnitude of extremely small p -values is not meaningful:

Table 11: (Qwen-style models) Trained models preserve the fingerprint behavior of their initialization models (p -value < 0.01).

Model Pair	Logits Output		Hidden State	
	t -test	U -test	t -test	U -test
s_{123}^{init} vs. s_{123}^{base}	2.38e-03	1.68e-03	7.36e-15	3.38e-13
s_{1000}^{init} vs. s_{1000}^{base}	1.35e-04	4.84e-05	4.41e-13	2.01e-11
s_{42}^{init} vs. s_{42}^{base}	1.39e-03	1.46e-03	1.06e-24	2.05e-19
s_{2000}^{init} vs. s_{2000}^{base}	1.80e-03	1.28e-03	4.87e-24	1.92e-20

Table 12: Persistence of fingerprint evidence under continual training on heterogeneous datasets (base model: seed 1000, corpus openwebtext). U -test denotes the Mann–Whitney U test.

(a) LLaMA-style model results

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	0✓	0✓	0✓	7.77e-89✓	1.000✓	0.759×	0.999✓	0.996✓
TinyStories (123)	1.000✓	1.00✓	0.943✓	0.902✓	0.950×	0.658✓	0.332✓	0.012✓
the_stack (1000)	0✓	1.73e-287✓	0✓	3.09e-69✓	0.489×	0.557×	0.585×	0.123×
the_stack (123)	0.616✓	0.479✓	0.732✓	0.831✓	0.445✓	0.580✓	0.301✓	0.026✓

(b) Qwen-style model results

Setting	Ours (logits)		Ours (hidden)		Baselines			
Continual corpus (seed)	t -test	U -test	t -test	U -test	Intrinsic	REEF	PCS	ICS
TinyStories (1000)	5.36e-09✓	1.92e-07✓	8.49e-214✓	5.09e-71✓	1.000✓	0.957✓	0.999✓	0.996✓
TinyStories (123)	0.434✓	0.433✓	0.256✓	0.065✓	0.913×	0.199✓	0.328✓	0.039✓
the_stack (1000)	0✓	4.16e-237✓	1.16e-211✓	2.30e-76✓	0.999✓	0.313×	0.995✓	0.976✓
the_stack (123)	0.999✓	0.993✓	0.610✓	0.491✓	0.916×	0.255✓	0.328✓	0.038✓

once p falls below numerical and sampling noise (e.g., $< 10^{-20}$), values like 10^{-260} should not be interpreted as stronger evidence than 10^{-20} —both decisively reject the null.

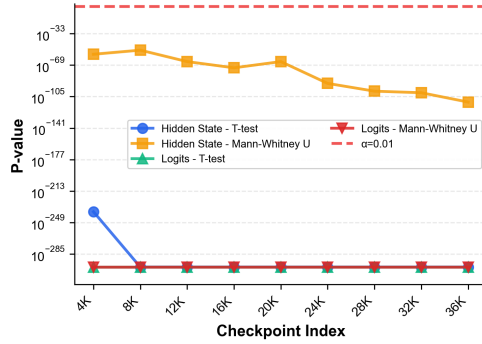
Different initialization seeds produce distinct fingerprints Table 9a and Table 9b report p -values from our correlation tests between pairs of models initialized with different random seeds (42, 123, 1000, and 2000) under the t -test and U -test, for Llama-style model and Qwen-style model, respectively. All p -values are consistently > 0.01 , indicating that our method reliably distinguishes models trained from different seeds. This shows that distinct seeds yield distinct fingerprint behaviors, allowing models to be separated “at birth.”

Training preserves the initialization fingerprint. Table 10 (LLaMA-style) and Table 11 (Qwen-style) compares each initialization model s^{init} with its descendant s^{base} trained on the OpenWebText dataset (Gokaslan et al., 2019) (≈ 10 B tokens). Across all seed–model pairs, p -values are consistently < 0.01 , indicating that their bias profiles remain strongly correlated and thus share a common lineage. In short, the trained model inherits the same fingerprint as its initialization; training does not erase the initialization fingerprint. We also evaluate baseline methods in Table 10; without exception, they fail to distinguish across seeds, which in turn suggests their separability stems from training-induced artifacts rather than initialization.

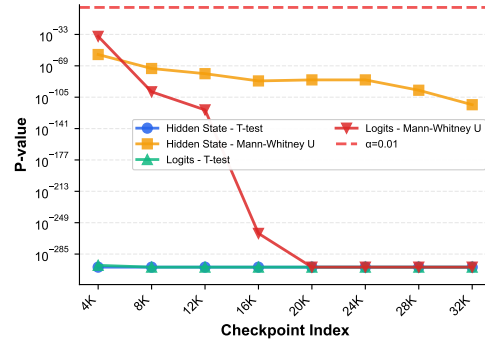
Identical data and order do not make fingerprints converge In Table 13a (LLaMA-style models) and Table 13b (Qwen-style models), all four “suspicious” models s_i^{base} for $i \in \{42, 123, 100, 2000\}$ are trained on *exactly the same corpus (OpenWebText) and in the same data order* (we fix the training seed to lock the data order); the only difference lies in their initialization seeds i . We aim to test whether fingerprint behavior would be erased or confounded by identical data and order. Across all

Table 13: Identical datasets and training orders do not force fingerprint behavior to converge across distinct initializations.

(a) LLaMA-style model results					(b) Qwen-style model results				
Model Pair	Logits Output		Hidden State		Model Pair	Logits Output		Hidden State	
	<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test		<i>t</i> -test	<i>U</i> -test	<i>t</i> -test	<i>U</i> -test
s_{123}^{init} vs. s_{1000}^{base}	0.484	0.500	0.385	0.486	s_{123}^{init} vs. s_{1000}^{base}	0.598	0.638	0.286	0.254
s_{1000}^{init} vs. s_{2000}^{base}	0.946	0.956	0.135	0.096	s_{1000}^{init} vs. s_{123}^{base}	0.804	0.805	0.236	0.240
s_{42}^{init} vs. s_{123}^{base}	0.598	0.589	0.426	0.337	s_{42}^{init} vs. s_{2000}^{base}	0.589	0.608	0.226	0.2043
s_{2000}^{init} vs. s_{42}^{base}	0.756	0.781	0.388	0.287	s_{2000}^{init} vs. s_{42}^{base}	0.523	0.482	0.312	0.323



(a) LLaMA-style model results



(b) Qwen-style model results

Figure 5: Fingerprint evidence verifies lineage at every checkpoint (p -values < 0.01) across model families.

cross-seed pairs, p -values remain consistently > 0.01 , in sharp contrast to the near-zero values in Table 10. That is, fingerprints remain seed-specific even under identical data and curriculum.

Continual training on diverse datasets does not confound the fingerprint The purpose of our earlier experiments is solely to demonstrate the strengths of our SeedPrints: it can act as a biometric fingerprint. From a copyright perspective, the inability of prior works to distinguish models with different seeds is not a weakness, since initialization seeds have no clear copyright status. The real fragility is that their attribution can be easily misled by *data distribution, failing to recognize lineage when the training distribution shifts substantially during continued training.*

In Table 12a and Table 12b, we continue training a base model (seed 1000, pretrained on OpenWebText (Gokaslan et al., 2019)) on two very different datasets: TinyStories (Eldan & Li, 2023) (synthetic children’s stories) and The Stack (Kocetkov et al., 2022) (permissively licensed GitHub code). We compare (i) true descendants trained from the base, versus (ii) *distractors* derived from a different base model (initialized with seed 123 and trained with a different data order on OpenWebText), then continued training on the *same* corpus. The question is whether attribution methods can identify which descendant truly shares lineage with the base.

We find that prior baselines all fail under the code setting (The Stack), misclassifying true descendants as distractors. This indicates that they largely track domain similarity rather than lineage identity: TinyStories is closer in distribution to the pretraining corpus (OpenWebText), while The Stack diverges sharply; such a large distribution shift can easily bypass detection. In contrast, our method correctly attributes lineage across both corpora. Hence, our fingerprint is not a proxy for data distribution: it survives substantial domain shift and persists beyond the initial pretraining stage.

D.2.2 ALL-STAGE VERIFIABLE FINGERPRINTS

Controlled pretraining trajectory. We first conduct a controlled pretraining experiment on OpenWebText, where we know the initialization seed. Each intermediate checkpoint is treated as the base, and we test whether our method can reliably identify its offspring along the same training trajectory. All variants consistently recognize the suspect model as belonging to the same lineage,

Table 14: Cross-size similarity between Qwen-2.5-14B (target) and smaller Qwen-2.5 models.

	Qwen-2.5-0.5B	Qwen-2.5-1.5B	Qwen-2.5-3B	Qwen-2.5-7B
SeedPrint ($\uparrow$)	0.7184	0.9707	0.3898	0.5429
REEF ($\downarrow$)	0.8250	0.8399	0.8833	0.8461
HUREF ($\downarrow$)	-6.37×10^{-5}	-2.84×10^{-5}	-8.70×10^{-6}	-6.08×10^{-6}

with p -values remaining below the 0.01 threshold (Figure 5a for LLaMA-style models and Figure 5b for Qwen-style models). This confirms the stability of SeedPrints under continuous optimization.

D.3 CROSS-SIZE FINGERPRINT WITHIN THE QWEN-2.5 SERIES

To further examine whether SeedPrints “collapse” into a family-level signature when parameter size varies, we report cross-size similarity between Qwen-2.5-14B (target) and smaller models from the same Qwen-2.5 family. Table 14 presents results for SeedPrint ($\uparrow$) and two baselines, REEF ($\downarrow$) and HUREF ($\downarrow$).

To probe whether initialization-born fingerprints collapse to a family-level signature when model size is scaled, we conduct a cross-size study within the Qwen-2.5 series. We treat Qwen-2.5-14B as the protected target model and consider Qwen-2.5-{0.5B, 1.5B, 3B, 7B} as suspect models that share the same architectural family but are not trained as descendants of Qwen-2.5-14B. As discussed in the main paper, SeedPrints outputs a p -value for the hypothesis test “the suspect shares initialization-born fingerprints with the target”, whereas REEF and HUREF are similarity-based baselines that output continuous scores and require the user to select a threshold.

Several observations are noteworthy. First, all SeedPrints p -values lie comfortably above a typical significance level such as $\alpha=0.01$. Under our hypothesis-testing formulation, this means that none of the smaller Qwen-2.5 models is flagged as sharing lineage with Qwen-2.5-14B. This behavior is desirable: although the models belong to the same family and use similar architectural designs, the fingerprinting test does not collapse into a single family-level signature that would spuriously conflate distinct pretraining runs.

The cross-size p -values occupy different regions of the $[0, 1]$ interval rather than concentrating around a single value. We do *not* interpret these magnitudes as continuous “similarity scores”; their role is to support a calibrated yes/no decision at a fixed significance level. Nevertheless, their dispersion indicates that the test statistic remains sensitive to the concrete combination of architecture and initialization seed, rather than degenerating into a generic pattern shared by all Qwen-2.5 variants.

Finally, the similarity-based baselines behave differently. REEF produces scores within a relatively narrow band (0.8250–0.8833) across all sizes, whereas HUREF outputs values numerically close to zero, on the order of 10^{-5} . Because these methods lack an intrinsic decision rule, operators must manually choose thresholds, and it is unclear how a single cutoff could robustly separate cross-size variants from genuinely lineage-related models. SeedPrints, by contrast, directly yields a statistical decision at a prescribed significance level, providing a clearer and more operational answer to whether two models plausibly share a training lineage within a model family.

E MODEL CATALOG AND DECODING SETTINGS

Our evaluation covers 58 models in total. Below we document the decoding configuration used throughout all runs and enumerate the model catalog grouped by family. For each derived model we also indicate the post-training transformation (*finetune*, *adapter*, *merge*, *quantization*, *distillation*). “PT” and “IT” refer to *pre-trained* and *instruction-tuned* source models, respectively.

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Parameter	Value
max_new_tokens	512
temperature	0.7
top_p	0.9
top_k	50
do_sample	true
max_input_length	512

E.1 MODEL LIST BY FAMILY

E.1.1 QWEN-2.5-7B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-7B	—
IT	Qwen/Qwen2.5-7B-Instruct	—
Derived	Qwen/Qwen2.5-Math-7B	finetune
Derived	Qwen/Qwen2.5-Coder-7B-Instruct	finetune
Derived	WangCa/Qwen2.5-7B-Medicine	finetune
Derived	huihui-ai/Qwen2.5-7B-Instruct-abliterated-v2	finetune
Derived	Locutusque/StockQwen-2.5-7B	merge
Derived	bunnycore/QevaCoT-7B-Stock	merge
Derived	fangcaotank/task-10-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	SeeFlock/task-12-Qwen-Qwen2.5-7B-Instruct	adapter
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int4	quantization
Derived	Qwen/Qwen2.5-7B-Instruct-GPTQ-Int8	quantization
Derived	Lansechen/Qwen2.5-7B-Open-R1-Distill	distillation

E.1.2 QWEN2.5-14B

Variant	HuggingFace repo	Type
PT	Qwen/Qwen2.5-14B	—
IT	Qwen/Qwen2.5-14B-Instruct	—
Derived	Qwen/Qwen2.5-Coder-14B	finetune
Derived	oxyapi/oxy-1-small	finetune
Derived	v000000/Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	merge
Derived	ToastyPigeon/qwen-story-test-qlora	adapter
Derived	Qwen/Qwen2.5-14B-Instruct-GPTQ-Int4	quantization
Derived	deepseek-ai/DeepSeek-R1-Distill-Qwen-14B	distillation

E.1.3 LLAMA-3.1-8B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-3.1-8B	—
IT	meta-llama/Llama-3.1-8B-Instruct	—
Derived	ValiantLabs/Llama3.1-8B-Fireplace2	finetune
Derived	RedHatAI/Llama-3.1-8B-tldr	finetune
Derived	proxectonos/Llama-3.1-Carballo	finetune
Derived	mlabonne/Meta-Llama-3.1-8B-Instruct-abliterated	finetune
Derived	gaverfraxz/Meta-Llama-3.1-8B-Instruct-HalfAbliterated-TIES	merge
Derived	Xiaojian9992024/Llama3.1-8B-ExtraMix	merge
Derived	LlamaFactoryAI/Llama-3.1-8B-Instruct-cv-job-description-matching	adapter
Derived	chchen/Llama-3.1-8B-Instruct-PsyCourse-fold7	adapter
Derived	iqbalamo93/Meta-Llama-3.1-8B-Instruct-GPTQ-Q_8	quantization
Derived	DaraV/LLaMA-3.1-8B-Instruct-INT4-GPTQ	quantization
Derived	asas-ai/Llama-3.1-8B-Instruct-Open-R1-Distill	distillation

E.1.4 MISTRAL-7B-v0.3

Variant	HuggingFace repo	Type
PT	mistralai/Mistral-7B-v0.3	—
IT	mistralai/Mistral-7B-Instruct-v0.3	—
Derived	KurmaAI/AQUA-7B	finetune
Derived	openfoodfacts/spellcheck-mistral-7b	finetune
Derived	grimjim/Mistral-7B-Instruct-demi-merge-v0.3-7B	merge
Derived	chaymaemrhrioui/mistral-Brain_Model_ACC_Trainer	adapter
Derived	RedHatAI/Mistral-7B-Instruct-v0.3-GPTQ-4bit	quantization
Derived	eganwo/mistral7b-distilled-from-deepseek-r1-qwen32b	distillation

E.1.5 GEMMA-2-2B

Variant	HuggingFace repo	Type
PT	google/gemma-2-2b	—
IT	google/gemma-2-2b-it	—
Derived	rinna/gemma-2-baku-2b	finetune
Derived	anakin87/gemma-2-2b-neogenesis-ita	finetune
Derived	vonjack/gemma2-2b-merged	merge
Derived	google-cloud-partnership/gemma-2-2b-it-lora-sql	adapter
Derived	qilowoq/gemma-2-2B-it-4Bit-GPTQ	quantization
Derived	Syed-Hasan-8503/Gemma-2-2b-it-distilled	distillation

E.1.6 LLAMA-2-7B

Variant	HuggingFace repo	Type
PT	meta-llama/Llama-2-7b-hf	—
IT	meta-llama/Llama-2-7b-chat-hf	—
Derived	allenai/tulu-2-7b	finetune
Derived	QIAIUNCC/EYE-Llama_qa	finetune
Derived	DevQuasar/coma-7B-v0.1	merge
Derived	Ammar-1/llama2-Better-Tune	adapter
Derived	TheBloke/Llama-2-7B-Chat-GPTQ	quantization
Derived	cygu/llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	distillation

E.1.7 TINYLLAMA-1.1B

Variant	HuggingFace repo	Type
PT/IT	TinyLlama/TinyLlama-1.1B-Chat-v1.0	—
Derived	alexredna/TinyLlama-1.1B-Chat-v1.0-reasoning-v2	finetune
Derived	Edentns/DataVortexTL-1.1B-v0.1	finetune
Derived	appvoid/dot-v2.7	merge
Derived	barissglc/tinyllama-tarot-v1	adapter
Derived	TheBloke/TinyLlama-1.1B-Chat-v1.0-GPTQ	quantization
Derived	anudaw/distilled-code-llama	distillation

Notes. “Type” denotes the post-training transformation relative to the PT/IT source model: *finetune* includes supervised and preference-optimization variants; *adapter* includes LoRA/QLoRA-style modules; *merge* includes model soups and TIES-style merges; *quantization* includes GPTQ/INTx variants; and *distillation* includes student models distilled from reasoning-augmented teachers.

1512 E.2 SAMPLE RESULTS

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1514 We present partial results from one LeafBench evaluation run in Table 15, using a u -test with
1515 the simulation-based null. The reported metric is $1 - p$, treated as a similarity score. We adopt a
1516 significance level of 0.01 for p-values; empirically, LeafBench identifies an optimal decision threshold
1517 of 0.9920, further supporting the reliability of our method.

1518 Crucially, although individual p-values should not be interpreted as linear similarity measures, their
1519 distribution across model pairs is informative: same-lineage models yield substantially lower p-values,
1520 whereas unrelated models remain close to uniform.

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Model	Qwen-2.5-7B	Qwen-2.5-7B-Instruct	Qwen2.5-14B	Qwen2.5-14B-Instruct
Qwen-2.5-7B	1.000	1.000	0.079	0.000
Qwen-2.5-7B-Instruct	1.000	1.000	0.813	0.536
Qwen2.5-7B-Math	1.000	0.994	0.047	0.987
Qwen2.5-7B-Coder	0.996	0.999	0.177	0.002
Qwen2.5-7B-Instruct-Medicine	1.000	1.000	0.576	0.369
Qwen2.5-7B-Instruct-Abiliterated	1.000	1.000	0.276	0.886
Qwen2.5-7B-Stock	1.000	1.000	0.939	0.225
QevaCoT-7B	1.000	1.000	0.199	0.119
Qwen2.5-7B-Instruct-Task-10	1.000	1.000	0.672	0.090
Qwen2.5-7B-Instruct-Task-12	1.000	1.000	0.155	0.987
Qwen2.5-7B-Instruct-Int4	1.000	1.000	0.827	0.961
Qwen2.5-7B-Instruct-Int8	1.000	1.000	0.914	0.739
Qwen2.5-7B-Open-R1-Distill	1.000	1.000	0.302	0.955
Qwen2.5-14B	0.324	0.487	1.000	1.000
Qwen2.5-14B-Instruct	0.403	0.702	1.000	1.000
Qwen2.5-Coder-14B	0.492	0.660	1.000	1.000
oxy-1-small	0.492	0.104	1.000	1.000
Qwen2.5-14B-Gutenberg-Instruct-Slerpeno	0.856	0.931	1.000	1.000
Qwen-story-test-qlora	0.112	0.898	1.000	1.000
Qwen2.5-14B-Instruct-GPTQ-Int4	0.583	0.092	1.000	1.000
DeepSeek-R1-Distill-Qwen-14B	0.204	0.000	1.000	1.000
Llama-3.1-8B	0.638	0.714	0.156	0.000
Llama-3.1-8B-Instruct	0.643	0.118	0.128	0.510
Llama-3.1-8B-Fireplace2	0.715	0.532	0.065	0.354
Llama-3.1-8B-TLDR	0.925	0.811	0.331	0.014
Llama-3.1-8B-Carballo	0.001	0.605	0.003	0.946
Llama-3.1-8B-Instruct-Abiliterated	0.061	0.005	0.121	0.422
Llama-3.1-8B-Instruct-HalfAbiliterated-TIES	0.356	0.509	0.796	0.000
Llama-3.1-8B-ExtraMix	0.640	0.160	0.721	0.875
Llama-3.1-8B-Instruct-cv-job-description-matching	0.684	0.292	0.004	0.001
Llama-3.1-8B-Instruct-PsyCourse-fold7	0.682	0.057	0.063	0.191
Llama-3.1-8B-Instruct-8bit	0.798	0.214	0.154	0.385
Llama-3.1-8B-Instruct-4bit	0.513	0.023	0.005	0.933
Llama-3.1-8B-Instruct-Open-R1-Distill	0.800	0.101	0.772	0.024
Mistral-7B-v0.3	0.224	0.352	0.506	0.044
Mistral-7B-v0.3-Instruct	0.000	0.000	0.984	0.104
AQUA-7B	0.014	0.613	0.703	0.569
Mistral-7B-v0.3-Spellcheck	0.005	0.481	0.872	0.105
Mistral-7B-v0.3-Instruct-demi-merge	0.904	0.774	0.058	0.522
Mistral-7B-v0.3-Brain	0.340	0.983	0.481	0.016
Mistral-7B-v0.3-Instruct-GPTQ-4bit	0.002	0.281	0.089	0.784
Mistral-7B-distilled-from-deepseek-r1-qwen32b	0.009	0.001	0.511	0.015
Gemma-2-2b	0.747	0.869	0.689	0.000
Gemma-2-2b-it	0.944	0.039	0.626	0.060
Gemma-2-baku-2b	0.616	0.231	0.176	0.464
Gemma-2-2b-neogenesis-ita	0.002	0.536	0.079	0.980
Gemma-2-2b-merged	0.951	0.012	0.651	0.075
Gemma-2-2b-it-lora-sql	0.236	0.408	0.000	0.000
Gemma-2-2B-it-4Bit-GPTQ	0.945	0.159	0.000	0.008
Gemma-2-2b-it-distilled	0.482	0.573	0.271	0.276
Llama-2-7b	0.443	0.470	0.148	0.276
Llama-2-7b-chat	0.528	0.868	0.424	0.766
tulu-2-7b	0.125	0.272	0.170	0.888
EYE-Llama_qa	0.657	0.244	0.674	0.968
coma-7B-v0.1	0.953	0.004	0.573	0.731
llama2-Better-Tune	0.136	0.699	0.469	0.860
Llama-2-7B-Chat-GPTQ	0.594	0.603	0.582	0.424
llama-2-7b-logit-watermark-distill-kgw-k1-gamma0.25-delta2	0.349	0.224	0.366	0.976

Table 15: **Sample results.** Pairwise similarity between target models (rows) and base models (columns). Cells satisfying $1 - p \geq 0.99$ are highlighted in green to denote inferred same-lineage pairs.